GreenTwin digital twin: baseline, what-if and retrofit worked examples

Part 1 · Household inputs → initial distributions → JAX physics → output distributions Team MegaMind · 11 October 2026 Model: thermal-electricity-fuels/0.3.0 · Worktree: PHYSICS MODEL AND BAYESIAN

This document explains exactly how the implemented physics block handles a household, a hypothetical change and a retrofit package. It follows the Part 1 separation in the system design: prepare uncertain inputs separately, run deterministic physics for each realization, and return distributions and joint samples for the next parts. The examples use saved, reproducible simulation runs. All ranges are **prior predictive estimates**, with Bayesian meter calibration reserved for the next phase.

1. The three cases use one physics block

Case	What the caller supplies	What happens inside the block	What the caller receives
A. Baseline household	Recorded property and customer answers	Preserve supplied facts; construct distributions for unknown quantities; sample and simulate	Baseline energy, fuel, heat-loss and electricity distributions
B. User what-if	The baseline household plus proposed changed facts	Prepare the changed household and rerun the same solver	After-state distributions and signed per-fuel changes against baseline
C. Retrofit package	The baseline household plus the package's physical edits	Apply all edits together and rerun the coupled thermal, fuel and electricity calculation	Package distributions, interactions and signed per-fuel changes

The simulator responds to the physical state presented to it. It does not need to know whether the proposed state came from a customer, chat or the retrofit engine. This makes it a separate, reusable computation block: the same building and equipment definitions drive every case.

Its three stages have distinct responsibilities:

1. **Initial prior construction** chooses probability distributions for unknown constants and household quantities from recorded property information, eligible previous application answers and physical defaults.

2. **Missing-input generation** samples those distributions and constructs consistent geometry, demand, weather and electricity profiles. It produces a prepared input artifact separately from simulation.

3. **Deterministic JAX simulation** takes one complete realization at a time. Repeating this over the prepared samples produces the final distribution of outcomes.

For a fixed complete realization, the physics calculation is deterministic. Randomness enters during input preparation, before the solver.

2. The example household and evaluation settings

All three examples use the recorded SH2 showcase household, a 1970s cavity-wall semi-detached home. It is an example household, rather than a real customer record.

Input	Value used
Household identity	SH2
Location	BS16, South West England; latitude 51.487°, longitude −2.512°
Dwelling	House; semi-detached; built in 1972, age band 1967–1975
Total floor area	95 m² across 2 storeys
Wall description	Cavity wall, uninsulated
Roof description	Pitched loft, 100 mm loft insulation

Input	Value used
Floor description	Solid floor, uninsulated
Windows	Older double glazing, pre-2002
Baseline heating	Condensing mains-gas boiler
Baseline hot water	From the main heating system
Customer-confirmed occupants	4
Customer-confirmed thermostat	20°C
Example-specified cooking	Mains gas, 1 kWh/day
Cylinder	Explicitly absent
Existing solar PV and battery	Explicitly absent, 0 kWp and 0 kWh
Simulation period	January–December 2025, all 12 months
Random seed	42
Ensemble size	128 realizations per scenario
JAX batch size	8
Weather basis for these saved runs	Explicitly enabled development weather, sampled with uncertainty

The 1 kWh/day cooking value is an exact assumption for this demonstration. Ordinary customer intake can use a simple cooking-fuel/appliance answer, from which the generator constructs uncertain consumption.

Each realization has **17,520 half-hour electricity intervals** over this 365-day year. The four main evaluations—baseline, fabric-only, heat-pump-only and combined package—therefore contain **512 household realizations** and **8,970,240 electricity dispatch intervals**.

3. Exactly what happens when a baseline household is submitted

Step 1 — Validate the request and preserve recorded facts

The application passes a HouseholdRequest: household identity, property/handoff, customer inputs, requested months, seed and available weather or electricity data. A household without a property/handoff needs the basic location, total floor area, dwelling type and heating-system facts to describe the simulation.

Customer-provided facts remain fixed across realizations. Here, every realization has four occupants, a 20°C setpoint, 95 m² total dwelling area and the same chosen heating system. false means confirmed absence, 0 is a supplied numerical value, and a missing value is treated as unknown. Supplied physical measurements also remain exact within the requested calculation.

Step 2 — Read eligible previous application answers

For application requests, prepare_application_inputs reads the household-history snapshot before recording the current baseline. It selects relevant previous answers by available district, region, home type, floor area, construction age and equipment. Demand cohorts prefer a floor area within 30% where suitable records exist, and daily energy evidence can additionally match occupant count.

Each distinct household contributes its latest eligible baseline revision. The current household/customer group, future records and held-out evaluation records are excluded from the population evidence. Proposed what-ifs and retrofits do not become observations. Generated samples and simulator results are never treated as customer answers.

Recent answers receive more weight: recency weight = exp(−age in years / 10). With effective evidence weight N, the numeric pooling weight is N / (N + 20). For illustration, 20 equally weighted observations give a 50% pooling weight; 80 give 80%. Categorical distributions use twenty fallback pseudo-observations. These examples explain the rule; the SH2

evidence runs below use their saved property/default distributions and do not learn from real customer history.

The new household's own answers are used immediately as exact facts. After successful baseline input generation, eligible recorded facts can inform later households. Prior construction records the snapshot identity and cohort details for replay.

Step 3 — Construct initial distributions for constants and unknown inputs

The prior builder creates source-labelled, physically bounded marginals. A known value is exact; an unknown value receives a distribution appropriate to its role.

Unknown quantity	Initial distribution used by the implementation
Fabric U-values	Positive lognormal, initial log-SD 0.22
Ventilation air changes	Lognormal, initial log-SD 0.25; 0.1–3 per hour
Thermal bridging coefficient	Lognormal, initial log-SD 0.25; 0.001–0.5 W/m²K
Combustion/district efficiency	Scaled beta, initial concentration 45; bounded 0.1–1
Heat-pump seasonal COP	Lognormal, initial log-SD 0.12; bounded 1–7
Hot-water efficiency/COP	Equipment-dependent beta or lognormal; direct electric efficiency is 1
Appliance, lighting and water demand factors	Lognormal, initial log-SDs 0.25, 0.20 and 0.22
Geometry detail	Positive bounded factors coupled to the dwelling dimensions
Missing occupant/presence/equipment choices	Categorical probabilities
Pump/fan electrical power	Lognormal; 0.005–0.2 kW
Water temperature rise	Bounded normal, initial 37 ± 3 K; 20–55 K
Development monthly outdoor temperature	Bounded normal, initial SD 1.5°C
Development monthly solar irradiation	Lognormal, initial log-SD 0.15
Battery usable fraction and efficiencies	Scaled beta, each bounded below 1

Application evidence can update the initial distributions while retaining their physical domains and heterogeneity. This is population-informed initialization. Meter calibration is a separate later operation.

Step 4 — Generate 128 consistent complete inputs

The missing-input generator draws one instance of each unknown quantity for every realization. Named random streams make regeneration repeatable and preserve latent draws for unaffected variables across comparable scenarios.

The inputs are generated jointly where physics requires a dependency. The footprint follows total area and storeys; shared geometry factors determine envelope areas and volume. Appliance electricity and its retained internal heat use the same realization. Hot-water demand follows occupants, water use and water heating energy. Monthly demand/weather variation preserves temporal dependence; the monthly profile copula uses AR(1) correlation 0.65.

The prepared artifact contains both distribution summaries and the actual aligned joint draws. The simulator uses those draws, rather than independently sampling the summaries again.

Step 5 — Calculate heat loss and useful heating with JAX

For each realization, the heat-loss coefficient is:

$H = \sum(U \times \text{exposed area}) + \text{thermal bridging coefficient} \times \text{exposed area} + 0.33 \times \text{ACH} \times \text{volume}$

H is in W/K. The monthly steady-state resistance calculation combines outdoor temperature, occupied heating hours, the setpoint, a 3 K setback balance, internal gains and solar gains. Useful heating is clipped at zero, with unused gains explicitly accounted for. This calculates monthly heat demand for the chosen household state.

Step 6 — Convert useful heat into purchased energy

Space heating and hot water are calculated separately using their selected equipment efficiency or COP:

purchased heating energy = useful heating / efficiency or COP

For combustion, the balance includes useful heat and conversion loss. For a heat pump, the balance includes purchased electricity and heat collected from the environment. Cooking, boiler auxiliaries, secondary heating, electric showers and cylinder losses follow their respective end uses. Electric-shower demand is separated from the main hot-water load to avoid double counting.

Supported purchased carriers are **grid electricity, mains gas, oil, LPG, solid fuel and district heat**. The gas/fuel calculation returns energy in kWh, allocated by carrier and end use.

Step 7 — Run half-hour electricity and storage dispatch

The realization's baseload, electric heating/hot water, auxiliaries and other electrical end uses are placed into half-hour profiles. The electricity engine balances demand, PV generation, direct solar consumption, grid import, export, battery charging/discharging, storage losses, curtailment and state of charge. Device power, usable-capacity and export limits apply.

PV reduces imported electricity. It does not erase the electrical demand being served. Battery state carries continuously through the simulated intervals.

Step 8 — Check physical accounting for every realization

The wrapper validates thermal balances, per-fuel conversion, electrical end uses, interval dispatch, battery storage and monthly aggregation. It records the maximum residual for each checked identity in max_conservation_error_kwh.

Step 9 — Summarize the 128 outcomes as distributions

For output draws $y_1 \ldots y_n$, the reported mean is $\sum y / n$; sample variance is $\sum (y - \text{mean})^2 / (n - 1)$. SD is the square root of variance. The Monte Carlo standard error of the mean is SD / √n.

Each numeric distribution also carries median, mode, percentiles, 50% and 90% intervals, skewness, a histogram and sample count. Here the 90% range is P5–P95 of the prior predictive realizations. Monte Carlo standard error measures sampling precision of the mean; SD describes spread across the modeled household realizations.

Step 10 — Return and save the result

The caller receives EnsembleResult plus aligned joint output samples. Preparation and simulation can be run separately: prepared inputs are hash-verified before replay, and results retain identity, seed, versions, input provenance and calibration status.

4. Case A — A baseline household: one realization and the final result

The first baseline realization shows the calculation concretely. Its setpoint is exactly 20°C. Sampled heating duration is 6.621636 hours/day, boiler efficiency is 0.869889 and hot-water efficiency is 0.947163.

January 2025 calculation, realization 0	Result
Heat-loss coefficient	313.09338 W/K
Heat loss before gains	3,032.84619 kWh
Useful heat gains	501.03958 kWh
Useful space heating	2,531.80640 kWh
Purchased gas for space heating	2,910.49341 kWh

January 2025 calculation, realization 0	Result
Purchased gas for hot water	183.13313 kWh
Gas cooking, 31 days × 1 kWh/day	31.00000 kWh
Total gas	3,124.62646 kWh
Grid electricity	371.63287 kWh

The path is visible in the arithmetic: 3,032.84619 − 501.03958 ≈ 2,531.80640 kWh of useful heating; 2,531.80640 / 0.869889 ≈ 2,910.49341 kWh of heating gas; and 2,910.49341 + 183.13313 + 31 ≈ 3,124.62646 kWh total gas. Small rounding differences come from displayed precision and numerical arithmetic.

This is one complete realization. The final annual estimate summarizes all 128 realizations:

Baseline annual output	Mean, kWh	P5–P95, kWh
Useful space heating	13,178.77	8,797.39–17,192.33
Useful hot water	2,143.63	1,431.35–2,914.89
Purchased mains gas	18,705.97	13,226.75–24,269.85
Grid electricity import	4,307.82	3,081.04–5,701.40
All-fuel purchased energy	23,013.79	17,843.19–28,973.26

Baseline gas variance is **11,689,846.14 kWh²**, its SD is **3,419.04 kWh**, and the Monte Carlo standard error of its mean is **302.20 kWh**. All-fuel purchased-energy variance is **11,588,959.41 kWh²**, with SD **3,404.26 kWh**.

For this household, all-fuel purchased energy equals grid import plus gas because oil, LPG, solid fuel and district heat are absent. PV generation and export are exactly zero because existing PV absence is explicit.

5. Case B — A user asks “What if I replace the gas boiler with a heat pump?”

The proposed change for this example is heating_system = air_source_heat_pump. The scenario starts from the baseline household and retains the same envelope, occupants, thermostat, cooking choice, seed, months and sample count. Heating equipment and its efficiency/COP distribution are prepared for the proposed state; the solver reruns that entire state.

The example's hot water follows the main heating system, so both main heating and hot-water energy move to electricity. The cooking appliance remains gas-fired. For an application comparison, use the same population/weather snapshot for both states and mark the new request as a what-if. Hypothetical answers stay outside the recorded baseline population.

Annual output	Baseline mean, kWh	Heat-pump what-if mean, kWh
Useful space heating	13,178.77	13,178.77
Useful hot water	2,143.63	2,143.63
Purchased mains gas	18,705.97	365.00
Grid electricity import	4,307.82	10,620.71
All-fuel purchased energy	23,013.79	10,985.71

The useful heat required by the home stays the same. The selected device changes the purchased energy needed to provide it. The remaining **365 kWh/year of gas** is exactly the supplied cooking assumption: 1 × 365.

The final heat-pump grid-import distribution has mean **10,620.71 kWh**, variance **2,328,223.49 kWh²**, SD **1,525.85 kWh** and P5–P95 **8,412.88–13,320.72 kWh**. Total purchased energy has P5–P95 **8,777.88–13,685.72 kWh**.

Savings use matched draws: saving[i] = baseline[i] − after[i]. The mean gas saving is **18,340.97 kWh/year**, electricity saving is **−6,312.89 kWh/year**, and total purchased-energy saving is **12,028.08 kWh/year**. The negative electricity

saving correctly expresses increased electricity use after electrification. Savings ranges come from the paired difference distribution, rather than subtracting unrelated percentile endpoints.

6. Case C — The retrofit engine submits a combined package

The retrofit caller describes the physical after-state. The example package specifies:

```
{
  "wall_u": 0.30,
  "roof_u": 0.16,
  "window_u": 1.40,
  "heating_system": "air_source_heat_pump",
  "pv_kwp": 3.5,
  "battery_kwh": 5.0
}
```

The U-values are W/m²K; PV capacity is kWp; nominal battery capacity is kWh. These supplied values define the package being evaluated. Other unspecified quantities retain their appropriate distributions. The caller provides a coherent proposed state, including replacing any superseded exact equipment facts and updating ownership fields when present.

The twin computes the package as one state:

1. Improved fabric changes heat loss and useful heating.

2. The heat pump supplies the resulting useful heat and hot water.

3. Its electricity becomes part of household electrical demand.

4. PV is allocated against that demand at each half hour.

5. The battery shifts available solar energy subject to power, capacity and loss constraints.

6. The result is compared with the matched baseline by fuel.

These interactions are calculated by rerunning the model. Separate retrofit savings are never added together.

Annual output	Baseline mean, kWh	Combined-package mean, kWh	Package P5–P95, kWh
Useful space heating	13,178.77	5,369.32	3,296.89–7,384.13
Useful hot water	2,143.63	2,143.63	1,431.35–2,914.89
Purchased mains gas	18,705.97	365.00	365.00–365.00
Electricity demand	4,307.82	7,447.02	5,929.95–9,224.55
Grid electricity import	4,307.82	4,602.06	3,141.88–6,492.91
PV generation	0.00	3,300.45	2,372.44–4,548.18
Electricity export	0.00	377.30	29.36–1,097.43
All-fuel purchased energy	23,013.79	4,967.06	3,506.88–6,857.91

Mean signed savings are **18,340.97 kWh/year gas**, **−294.23 kWh/year electricity**, and **18,046.73 kWh/year total purchased energy**. The total reduction is **78.42% of baseline mean purchased energy**, calculated as 100 × (23,013.791595 − 4,967.057247) / 23,013.791595. This is a ratio of the displayed scenario means; a downstream distribution of percentage improvement uses matched sample-wise ratios.

The package reduces mean useful space-heating demand by **59.26%** and mean H from **304.81 to 172.65 W/K**, a **43.36%** reduction. Average PV self-consumption is **89.58%**, average electrical self-sufficiency is **38.79%**, and average peak import is **2.338 kW**. The package's purchased-energy variance is **1,050,328.35 kWh²**, with SD **1,024.86 kWh**.

The heat-pump-only scenario imports 10,620.71 kWh/year, while the full package imports 4,602.06 kWh/year. Fabric improvements, solar generation and storage therefore materially alter the electricity result after electrification. The separate fabric-only run below makes the interaction visible.

7. What the returned output contains

The saved baseline result contains **16 period-total distributions**, **60 monthly output series**, **7 element heat-loss distributions**, **7 corresponding element-share distributions** and **8 metric slots**. Complete calendar years additionally receive annual summaries; partial requested periods remain period totals.

Output group	Content and use
Useful heat	Space heating and hot water, enabling comparison of thermal demand and equipment input
Purchased energy	Grid import, gas, oil, LPG, solid fuel and district heat; both grid-plus-gas and all-fuel totals
Fuel-by-end-use series	Space heating, hot water and other consumption by carrier
Thermal accounting	Heat loss, gains, unused gains, main/secondary heat, conversion losses and ambient heat
Electrical accounting	Demand, baseload, hot water, auxiliaries, secondary heat, outdoor loads and other electricity
Solar/storage accounting	Generation, direct consumption, charge/discharge, storage change/loss, export and curtailment
Envelope diagnostics	H, thermal bridges, walls, roof, floor, windows, doors, party walls and ventilation contributions
Electricity metrics	Peak import, 16:00–19:00 import share, PV self-consumption and electrical self-sufficiency
Traceability	Input/prior summaries, exact facts, sources, history/context audit, identity, seed, versions and conservation checks
Joint output samples	Aligned realizations for correlated downstream savings, carbon, bill and ranking calculations

An undefined ratio, such as PV self-consumption when the household has no PV, has an explicit null status. Exact absence yields zero energy for that absent system.

Part 2 can use paired per-fuel draws with its sourced emissions factors and tariffs, and compute improvement potential and package rankings. Part 3 can consume energy intensity, property state and rating information for the Green Home Score. The property-value component can use the resulting scenario state and ratings. The scene component can use the estimated element contributions. Solar generation, import/export and self-consumption are physical inputs to downstream renewable scoring.

The returned physics distributions provide a traceable numerical basis for these parts. The same sample index should be preserved when downstream calculations combine dependent outputs.

8. Reproducible execution and artifact handoff

The functions used by the block are:

Stage	Entry point
Standalone initial distributions	build_priors(request) in greentwin/twin/priors.py
Application history and input preparation	prepare_application_inputs(request, n_samples, history=..., kind=...)
Separate missing-input generation	generate_missing_inputs(request, n_samples, prior=...)
Deterministic ensemble simulation	simulate_ensemble(prepared_inputs, batch_size=8)
Signed baseline-minus-after summaries	paired_differences(baseline_run, after_run)
Prepared replay and saved results	save_inputs, load_inputs and save_result

From this worktree, a customer request can be prepared and simulated separately:

```
.venv/bin/python scripts/run_uncertain_twin.py prepare --request household.json --samples 128 --output
artifacts/household-inputs
```

```
.venv/bin/python scripts/run_uncertain_twin.py simulate --prepared artifacts/household-inputs --output
artifacts/household-result
```

Prepare writes request.json, priors.json, inputs.json, input_samples.npz and manifest.json. The manifest records SHA-256 hashes. Simulation verifies them and uses the frozen draws. Its output contains result.json, output_samples.npz and metrics.json.

For a proposed physical change:

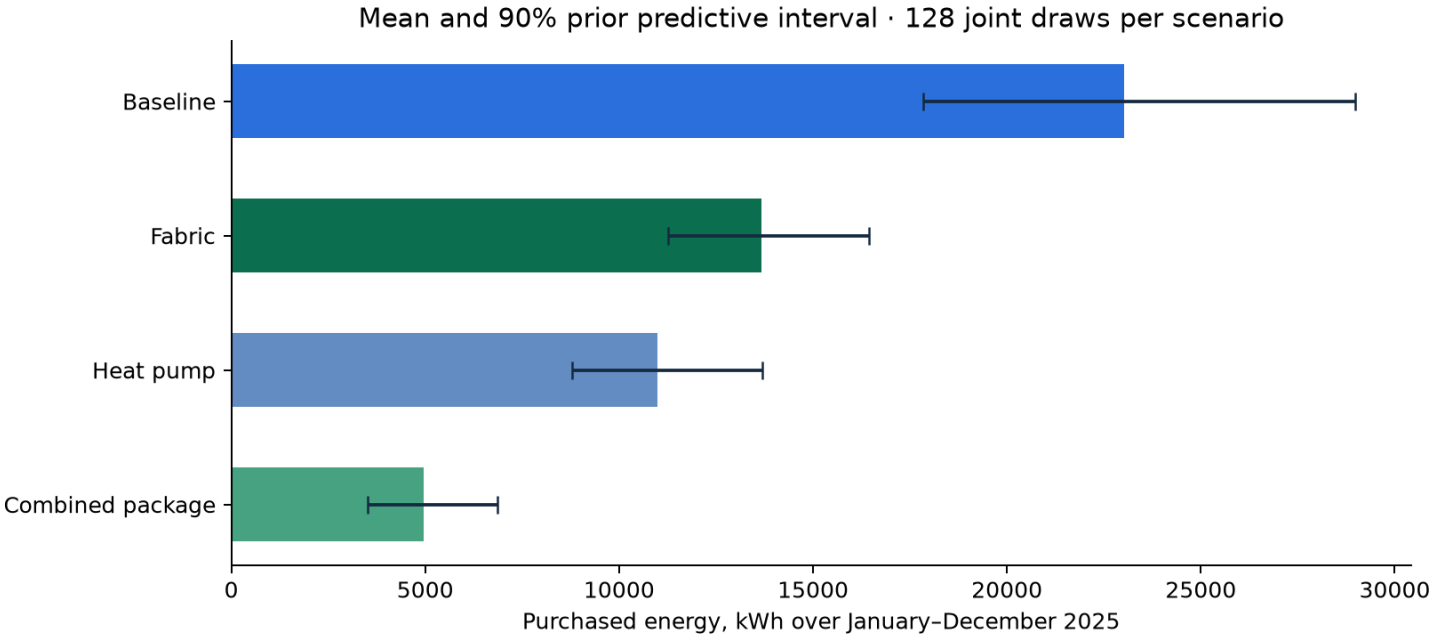
```
.venv/bin/python scripts/run_uncertain_twin.py prepare --request household.json --changes proposed-
changes.json --samples 128 --output artifacts/proposed-inputs
.venv/bin/python scripts/run_uncertain_twin.py simulate --prepared artifacts/proposed-inputs --output
artifacts/proposed-result
```

--changes is treated as a scenario, so it does not record proposed facts in baseline history. --what-if provides the same history behavior when passing an already modified request. For controlled comparisons, retain the same household, dates, seed, sample count and population/weather evidence; use named latent streams and compare aligned output draws. Each output directory is new and stays inside this worktree.

9. Evidence tables and complete output inventory

The tables below are extracted from the saved evaluation evidence and result artifacts. Means, variances, SDs and percentiles are reported directly from the simulations. Units are kWh for period energy, kWh² for energy variance, W/K for heat-loss coefficients, kW for peak power and percent for ratios. Display rounding does not change the stored full-precision values.

9.1 Scenario overview



Scenario	Gas mean	Grid-import mean	Purchased-energy mean	Purchased-energy P5–P95	Measured runtime
Baseline household	18,705.97	4,307.82	23,013.79	17,843.19–28,973.26	14.41 s
Fabric-only comparison	9,351.96	4,307.82	13,659.78	11,264.69–16,434.75	13.35 s
Heat-pump what-if	365.00	10,620.71	10,985.71	8,777.88–13,685.72	16.74 s
Combined retrofit package	365.00	4,602.06	4,967.06	3,506.88–6,857.91	12.66 s

Runtimes describe the saved local runs at 128 realizations and batch size 8, including their preparation/dispatch work within the simulation wrapper and any JAX compilation incurred there. They are measured run durations, rather than a

promised service latency.

9.2 Full period-total distribution summaries

Baseline household

Output	Mean	Variance	SD	P5	P95	MCSE
all_fuel_purchased_kwh	23,013.79	11,588,959.41	3,404.26	17,843.19	28,973.26	300.90
battery_final_kwh	0.00	0.00	0.00	0.00	0.00	0.00
battery_initial_kwh	0.00	0.00	0.00	0.00	0.00	0.00
battery_usable_capacity_kwh	0.00	0.00	0.00	0.00	0.00	0.00
district_kwh	0.00	0.00	0.00	0.00	0.00	0.00
electricity_demand_kwh	4,307.82	853,749.35	923.99	3,081.04	5,701.40	81.67
electricity_import_kwh	4,307.82	853,749.35	923.99	3,081.04	5,701.40	81.67
export_kwh	0.00	0.00	0.00	0.00	0.00	0.00
gas_kwh	18,705.97	11,689,846.14	3,419.04	13,226.75	24,269.85	302.20
hot_water_kwh	2,143.63	280,194.95	529.33	1,431.35	2,914.89	46.79
lpg_kwh	0.00	0.00	0.00	0.00	0.00	0.00
oil_kwh	0.00	0.00	0.00	0.00	0.00	0.00
purchased_electricity_gas_kwh	23,013.79	11,588,959.41	3,404.26	17,843.19	28,973.26	300.90
pv_generation_kwh	0.00	0.00	0.00	0.00	0.00	0.00
solid_kwh	0.00	0.00	0.00	0.00	0.00	0.00
space_heating_kwh	13,178.77	7,486,066.83	2,736.07	8,797.39	17,192.33	241.84

Fabric-only comparison

Output	Mean	Variance	SD	P5	P95	MCSE
all_fuel_purchased_kwh	13,659.78	2,955,395.39	1,719.13	11,264.69	16,434.75	151.95
battery_final_kwh	0.00	0.00	0.00	0.00	0.00	0.00
battery_initial_kwh	0.00	0.00	0.00	0.00	0.00	0.00
battery_usable_capacity_kwh	0.00	0.00	0.00	0.00	0.00	0.00
district_kwh	0.00	0.00	0.00	0.00	0.00	0.00
electricity_demand_kwh	4,307.82	853,749.35	923.99	3,081.04	5,701.40	81.67
electricity_import_kwh	4,307.82	853,749.35	923.99	3,081.04	5,701.40	81.67
export_kwh	0.00	0.00	0.00	0.00	0.00	0.00
gas_kwh	9,351.96	2,851,433.04	1,688.62	6,899.85	12,037.05	149.25
hot_water_kwh	2,143.63	280,194.95	529.33	1,431.35	2,914.89	46.79
lpg_kwh	0.00	0.00	0.00	0.00	0.00	0.00
oil_kwh	0.00	0.00	0.00	0.00	0.00	0.00
purchased_electricity_gas_kwh	13,659.78	2,955,395.39	1,719.13	11,264.69	16,434.75	151.95
pv_generation_kwh	0.00	0.00	0.00	0.00	0.00	0.00
solid_kwh	0.00	0.00	0.00	0.00	0.00	0.00

Output	Mean	Variance	SD	P5	P95	MCSE
space_heating_kwh	5,369.32	1,623,045.40	1,273.99	3,296.89	7,384.13	112.61

Heat-pump what-if

Output	Mean	Variance	SD	P5	P95	MCSE
all_fuel_purchased_kwh	10,985.71	2,328,223.63	1,525.85	8,777.88	13,685.72	134.87
battery_final_kwh	0.00	0.00	0.00	0.00	0.00	0.00
battery_initial_kwh	0.00	0.00	0.00	0.00	0.00	0.00
battery_usable_capacity_kwh	0.00	0.00	0.00	0.00	0.00	0.00
district_kwh	0.00	0.00	0.00	0.00	0.00	0.00
electricity_demand_kwh	10,620.71	2,328,223.49	1,525.85	8,412.88	13,320.72	134.87
electricity_import_kwh	10,620.71	2,328,223.49	1,525.85	8,412.88	13,320.72	134.87
export_kwh	0.00	0.00	0.00	0.00	0.00	0.00
gas_kwh	365.00	0.00	0.00	365.00	365.00	0.00
hot_water_kwh	2,143.63	280,194.95	529.33	1,431.35	2,914.89	46.79
lpg_kwh	0.00	0.00	0.00	0.00	0.00	0.00
oil_kwh	0.00	0.00	0.00	0.00	0.00	0.00
purchased_electricity_gas_kwh	10,985.71	2,328,223.63	1,525.85	8,777.88	13,685.72	134.87
pv_generation_kwh	0.00	0.00	0.00	0.00	0.00	0.00
solid_kwh	0.00	0.00	0.00	0.00	0.00	0.00
space_heating_kwh	13,178.77	7,486,066.83	2,736.07	8,797.39	17,192.33	241.84

Combined retrofit package

Output	Mean	Variance	SD	P5	P95	MCSE
all_fuel_purchased_kwh	4,967.06	1,050,328.35	1,024.86	3,506.88	6,857.91	90.59
battery_final_kwh	-0.00	0.00	0.00	0.00	0.00	0.00
battery_initial_kwh	0.00	0.00	0.00	0.00	0.00	0.00
battery_usable_capacity_kwh	4.50	0.01	0.09	4.36	4.65	0.01
district_kwh	0.00	0.00	0.00	0.00	0.00	0.00
electricity_demand_kwh	7,447.02	1,067,675.91	1,033.28	5,929.95	9,224.55	91.33
electricity_import_kwh	4,602.06	1,050,328.34	1,024.86	3,141.88	6,492.91	90.59
export_kwh	377.30	92,096.07	303.47	29.36	1,097.43	26.82
gas_kwh	365.00	0.00	0.00	365.00	365.00	0.00
hot_water_kwh	2,143.63	280,194.95	529.33	1,431.35	2,914.89	46.79
lpg_kwh	0.00	0.00	0.00	0.00	0.00	0.00
oil_kwh	0.00	0.00	0.00	0.00	0.00	0.00
purchased_electricity_gas_kwh	4,967.06	1,050,328.35	1,024.86	3,506.88	6,857.91	90.59
pv_generation_kwh	3,300.45	368,513.13	607.05	2,372.44	4,548.18	53.66
solid_kwh	0.00	0.00	0.00	0.00	0.00	0.00

Output	Mean	Variance	SD	P5	P95	MCSE
space_heating_kwh	5,369.32	1,623,045.40	1,273.99	3,296.89	7,384.13	112.61

9.3 Paired savings distributions

Positive means a reduction; negative means an increase. All rows use baseline minus the scenario for matched realizations.

Fabric-only comparison

Carrier / purchased-energy convention	Mean saving	Variance	SD	P5 saving	P95 saving
electricity_import_kwh	0.00	0.00	0.00	0.00	0.00
gas_kwh	9,354.01	6,416,256.49	2,533.03	6,010.84	13,918.47
oil_kwh	0.00	0.00	0.00	0.00	0.00
lpg_kwh	0.00	0.00	0.00	0.00	0.00
solid_kwh	0.00	0.00	0.00	0.00	0.00
district_kwh	0.00	0.00	0.00	0.00	0.00
purchased_electricity_gas_kwh	9,354.01	6,416,256.63	2,533.03	6,010.84	13,918.47
all_fuel_purchased_kwh	9,354.01	6,416,256.63	2,533.03	6,010.84	13,918.47

Heat-pump what-if

Carrier / purchased-energy convention	Mean saving	Variance	SD	P5 saving	P95 saving
electricity_import_kwh	-6,312.89	1,733,493.59	1,316.62	-8,504.59	-4,323.48
gas_kwh	18,340.97	11,689,846.14	3,419.04	12,861.75	23,904.85
oil_kwh	0.00	0.00	0.00	0.00	0.00
lpg_kwh	0.00	0.00	0.00	0.00	0.00
solid_kwh	0.00	0.00	0.00	0.00	0.00
district_kwh	0.00	0.00	0.00	0.00	0.00
purchased_electricity_gas_kwh	12,028.08	4,763,980.17	2,182.65	8,561.16	15,395.73
all_fuel_purchased_kwh	12,028.08	4,763,980.17	2,182.65	8,561.16	15,395.73

Combined retrofit package

Carrier / purchased-energy convention	Mean saving	Variance	SD	P5 saving	P95 saving
electricity_import_kwh	-294.23	618,351.87	786.35	-1,537.50	799.11
gas_kwh	18,340.97	11,689,846.14	3,419.04	12,861.75	23,904.85
oil_kwh	0.00	0.00	0.00	0.00	0.00
lpg_kwh	0.00	0.00	0.00	0.00	0.00
solid_kwh	0.00	0.00	0.00	0.00	0.00
district_kwh	0.00	0.00	0.00	0.00	0.00
purchased_electricity_gas_kwh	18,046.73	9,139,442.22	3,023.15	13,379.00	23,400.64
all_fuel_purchased_kwh	18,046.73	9,139,442.22	3,023.15	13,379.00	23,400.64

9.4 All metric summaries

Scenario	Metric	Mean	SD	P5	P95
Baseline household	peak_import_kw	1.4572	0.3138	1.0377	1.9624
Baseline household	peak_import_share_pct	21.4795	0.5197	20.6459	22.3564
Baseline household	self_consumption_pct	null: no PV denominator	—	—	—
Baseline household	self_sufficiency_pct	0.0000	0.0000	0.0000	0.0000
Baseline household	battery_initial_kwh	0.0000	0.0000	0.0000	0.0000
Baseline household	battery_final_kwh	0.0000	0.0000	0.0000	0.0000
Baseline household	heat_loss_coefficient_w_per_k	304.8083	33.5412	252.6221	364.2975
Baseline household	thermal_bridging_w_per_k	29.0920	7.6119	19.4327	41.8808
Fabric-only comparison	peak_import_kw	1.4572	0.3138	1.0377	1.9624
Fabric-only comparison	peak_import_share_pct	21.4795	0.5197	20.6459	22.3564
Fabric-only comparison	self_consumption_pct	null: no PV denominator	—	—	—
Fabric-only comparison	self_sufficiency_pct	0.0000	0.0000	0.0000	0.0000
Fabric-only comparison	battery_initial_kwh	0.0000	0.0000	0.0000	0.0000
Fabric-only comparison	battery_final_kwh	0.0000	0.0000	0.0000	0.0000
Fabric-only comparison	heat_loss_coefficient_w_per_k	172.6477	15.2500	148.7663	197.7367
Fabric-only comparison	thermal_bridging_w_per_k	29.0920	7.6119	19.4327	41.8808
Heat-pump what-if	peak_import_kw	3.2357	0.4468	2.6232	3.9486
Heat-pump what-if	peak_import_share_pct	16.8048	0.6393	15.7798	17.7948
Heat-pump what-if	self_consumption_pct	null: no PV denominator	—	—	—
Heat-pump what-if	self_sufficiency_pct	0.0000	0.0000	0.0000	0.0000
Heat-pump what-if	battery_initial_kwh	0.0000	0.0000	0.0000	0.0000
Heat-pump what-if	battery_final_kwh	0.0000	0.0000	0.0000	0.0000
Heat-pump what-if	heat_loss_coefficient_w_per_k	304.8083	33.5412	252.6221	364.2975
Heat-pump what-if	thermal_bridging_w_per_k	29.0920	7.6119	19.4327	41.8808
Combined retrofit package	peak_import_kw	2.3378	0.3145	1.8556	2.8697
Combined retrofit package	peak_import_share_pct	15.4041	1.6466	13.2731	18.2586
Combined retrofit package	self_consumption_pct	89.5817	6.7266	76.1772	98.9297
Combined retrofit package	self_sufficiency_pct	38.7856	6.4801	28.8989	49.6439
Combined retrofit package	battery_initial_kwh	0.0000	0.0000	0.0000	0.0000
Combined retrofit package	battery_final_kwh	-0.0000	0.0000	0.0000	0.0000

Scenario	Metric	Mean	SD	P5	P95
Combined retrofit package	heat_loss_coefficient_w_per_k	172.6477	15.2500	148.7663	197.7367
Combined retrofit package	thermal_bridging_w_per_k	29.0920	7.6119	19.4327	41.8808

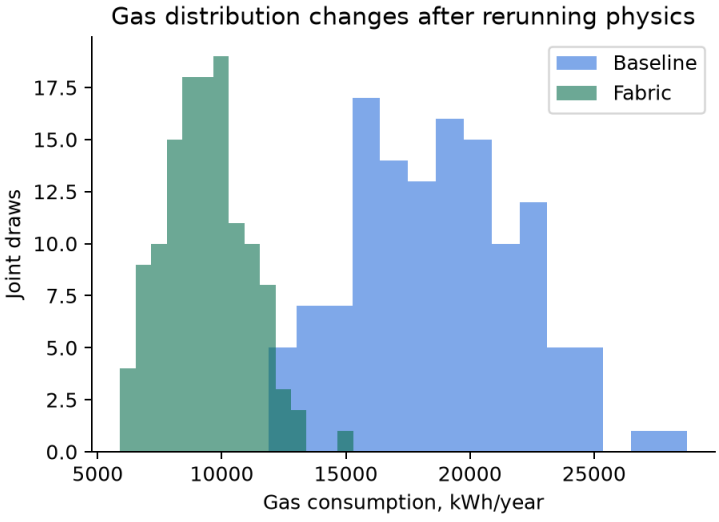
Self-consumption and self-sufficiency are means of realization-wise percentages. They need not equal a ratio formed from the mean energies. Battery usable capacity, initial/final storage and storage losses are available alongside these metrics.

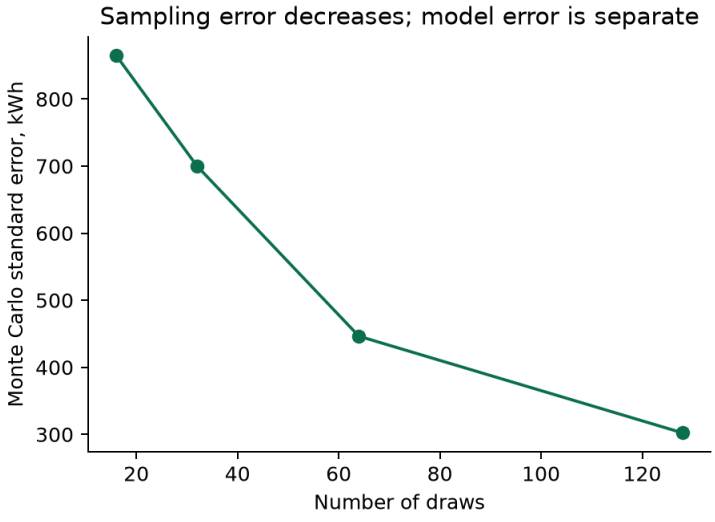
9.5 Physics, repeatability and uncertainty evidence

Check	Measured result
Fabric realizations with useful heating no higher than baseline	128/128
Identical regenerated parameter arrays for the same request and seed	True
Largest checked residual across four annual scenarios	0.00327297 kWh
H SD before seven explicit envelope measurements	33.54120 W/K
H SD after those measurements	3.93676 W/K
Reduction in H sample SD in that experiment	88.2629%

The seven exact envelope inputs in the information experiment are wall, roof, floor, window and door U-values, ventilation ACH and thermal bridging. It demonstrates how additional measured facts narrow the modeled uncertainty in H. Conservation residuals measure numerical accounting consistency. These numerical checks and scenario estimates are simulation evidence.

Sampling precision





Baseline draws included	Gas sample mean, kWh	MCSE of mean, kWh
16	18,743.01	865.07
32	19,083.27	699.97
64	19,111.28	446.36
128	18,705.97	302.20

The means above use prefixes of the same saved ensemble. Increasing the draw count improves sampling precision; the final customer range continues to represent uncertainty in the modeled inputs.

Six-carrier conversion checks

Each independent check uses eight January realizations with the same home and 1 kWh/day gas cooking. A non-gas heating system therefore retains 31 kWh of January gas cooking.

Heating carrier	Mean all-fuel purchased energy, January kWh	Maximum heating-conversion residual, kWh	Mean January gas, kWh
mains_gas	3,695.4989	0.00074111	3,347.0975
oil	3,867.3282	0.00066172	31.0000
lpg	3,695.4989	0.00074111	31.0000
solid	5,081.9278	0.00086164	31.0000
district	3,467.0948	0.00064432	31.0000
electricity	1,509.9398	0.00006691	31.0000

9.6 Complete monthly-series inventory

Each of these 60 fields has one output distribution per requested month. For a full year, that is 720 monthly distribution summaries per scenario. The complete numerical summaries are retained in the linked result JSON files.

Group	Exact output keys
Fuel and purchased energy	monthly_all_fuel_purchased_kwh
Fuel and purchased energy	monthly_district_hot_water_fuel_kwh
Fuel and purchased energy	monthly_district_kwh
Fuel and purchased energy	monthly_district_other_fuel_kwh
Fuel and purchased energy	monthly_district_space_heating_fuel_kwh
Fuel and purchased energy	monthly_electricity_hot_water_fuel_kwh
Fuel and purchased energy	monthly_electricity_other_fuel_kwh
Fuel and purchased energy	monthly_electricity_space_heating_fuel_kwh
Fuel and purchased energy	monthly_extra_gas_kwh
Fuel and purchased energy	monthly_gas_kwh
Fuel and purchased energy	monthly_heating_fuel_kwh
Fuel and purchased energy	monthly_hot_water_fuel_kwh
Fuel and purchased energy	monthly_lpg_hot_water_fuel_kwh
Fuel and purchased energy	monthly_lpg_kwh
Fuel and purchased energy	monthly_lpg_other_fuel_kwh
Fuel and purchased energy	monthly_lpg_space_heating_fuel_kwh
Fuel and purchased energy	monthly_mains_gas_hot_water_fuel_kwh
Fuel and purchased energy	monthly_mains_gas_other_fuel_kwh
Fuel and purchased energy	monthly_mains_gas_space_heating_fuel_kwh
Fuel and purchased energy	monthly_oil_hot_water_fuel_kwh
Fuel and purchased energy	monthly_oil_kwh
Fuel and purchased energy	monthly_oil_other_fuel_kwh
Fuel and purchased energy	monthly_oil_space_heating_fuel_kwh
Fuel and purchased energy	monthly_purchased_electricity_gas_kwh

Group	Exact output keys
Fuel and purchased energy	monthly_solid_hot_water_fuel_kwh
Fuel and purchased energy	monthly_solid_kwh
Fuel and purchased energy	monthly_solid_other_fuel_kwh
Fuel and purchased energy	monthly_solid_space_heating_fuel_kwh
Other monthly diagnostics	monthly_auxiliary_electricity_kwh
Other monthly diagnostics	monthly_baseload_kwh
Other monthly diagnostics	monthly_battery_charge_kwh
Other monthly diagnostics	monthly_battery_delta_kwh
Other monthly diagnostics	monthly_battery_discharge_kwh
Other monthly diagnostics	monthly_battery_loss_kwh
Other monthly diagnostics	monthly_curtailment_kwh
Other monthly diagnostics	monthly_direct_pv_kwh
Other monthly diagnostics	monthly_electricity_demand_kwh
Other monthly diagnostics	monthly_electricity_kwh
Other monthly diagnostics	monthly_export_kwh
Other monthly diagnostics	monthly_heat_gains_kwh
Other monthly diagnostics	monthly_heat_loss_kwh
Other monthly diagnostics	monthly_heating_ambient_heat_kwh
Other monthly diagnostics	monthly_heating_auxiliary_electricity_kwh
Other monthly diagnostics	monthly_heating_conversion_loss_kwh
Other monthly diagnostics	monthly_hot_water_ambient_heat_kwh
Other monthly diagnostics	monthly_hot_water_conversion_loss_kwh
Other monthly diagnostics	monthly_hot_water_kwh
Other monthly diagnostics	monthly_hot_water_system_loss_kwh
Other monthly diagnostics	monthly_internal_gains_kwh
Other monthly diagnostics	monthly_main_hot_water_kwh
Other monthly diagnostics	monthly_main_space_heating_kwh
Other monthly diagnostics	monthly_outdoor_other_electricity_kwh
Other monthly diagnostics	monthly_pv_generation_kwh
Other monthly diagnostics	monthly_secondary_electricity_kwh
Other monthly diagnostics	monthly_secondary_unused_request_kwh
Other monthly diagnostics	monthly_shower_electricity_kwh
Other monthly diagnostics	monthly_solar_gains_kwh
Other monthly diagnostics	monthly_space_heating_kwh
Other monthly diagnostics	monthly_thermal_storage_delta_kwh
Other monthly diagnostics	monthly_unused_heat_gains_kwh

9.7 Building-element diagnostics

Element	Baseline mean H contribution, W/K	Baseline mean share, %
doors	11.1206	3.6773
floor	36.1994	11.9633
party_walls	0.0000	0.0000
roof	25.6221	8.4724
ventilation	50.0837	16.4751
walls	138.4099	45.1303
windows	43.3727	14.2816

These contributions are estimates from the simulated building state. Their joint draws allow the scene to explain which elements contribute to heat loss.

9.8 Full-precision evidence and replay references

Evidence SHA-256: 7658a58c0c7dbf14d468559421c879a93fe214e882633ed31aee9c149d755973.

- Complete measured evaluation evidence
- Original prepared request
- Prepared priors
- Prepared input distributions
- Prepared joint inputs
- Prepared hash manifest
- Baseline household: full output distributions
- Baseline household: aligned output samples
- Fabric-only comparison: full output distributions
- Fabric-only comparison: aligned output samples
- Heat-pump what-if: full output distributions
- Heat-pump what-if: aligned output samples
- Combined retrofit package: full output distributions
- Combined retrofit package: aligned output samples
- Known-envelope experiment
- Application-history distribution method

The evidence files retain the complete distribution summaries, histograms, percentiles, monthly values and conservation checks beyond the displayed rounded tables. This document and those artifacts together make the baseline, what-if and retrofit examples inspectable and reproducible.